

A Proposal for a Modular Quantum Infrastructure: The NEPO
Project

"Foundational Architecture Proposal"

Armand J. Brunelle III

Independent Researcher

17 December 2025

Made in collaboration with; *AgentX (Copilot), GemX (Gemini) & Grok*

Acknowledgment of AI Assistance

This work was developed independently by the author with assistance from AI tools GemX, AgentX, and Grok in drafting and conceptualizing content. These tools contributed to organizing and refining ideas but do not constitute authorship.

Abstract

This work proposes a modular, hybrid quantum-classical infrastructure designed to address the structural limitations of contemporary communication networks in the approaching quantum era. Rather than treating quantum security as a standalone cryptographic upgrade, the proposed framework reframes secure communication as a convergent system problem involving hardware architecture, identity, memory, routing, and verification as co-dependent primitives.

The architecture introduces a layered, adaptive model in which quantum key distribution functions as an immutable security backbone, while classical computation, modular agents, and deterministic identity wrapping enable scalability, interoperability, and long-term evolution. Devices are treated not merely as endpoints, but as accountable participants within a distributed mesh, regardless of ownership or trust assumptions.

Central to the proposal is the notion of convergence across non-homogeneous systems: space-based infrastructure, terrestrial relays, and end-consumer devices coexist within a unified yet partitioned network topology. This work outlines the conceptual foundations of such a system, including asynchronous triangulation across space-time dimensions, modular quantum key management, identity abstraction via deterministic identifiers, and agent-assisted routing and memory orchestration.

The document is intentionally architectural rather than exhaustive. Formal mathematical derivations and physical implementations are deferred to supplementary sections in favor of clarity, modularity, and long-term applicability. The result is a foundational framework intended to remain adaptable as quantum technologies mature, rather than a fixed or brittle implementation tied to current hardware constraints.

Intended Audience and Scope

This document is written for multiple technical audiences with differing expectations, and it is not intended to serve as a step-by-step implementation guide.

- Systems architects and infrastructure designers will find the primary value in the modular network model, identity abstraction, memory partitioning, and agent-based routing concepts.
- Cryptographers and quantum information researchers may focus on the role of quantum key distribution as a structural primitive rather than a standalone protocol, and on the system-level implications of shared and partitioned quantum keys.
- Hardware and chip designers may engage with the layered microprocessor model, the Agent-on-Chip paradigm, and the discussion of material constraints and thermal behavior.
- Researchers in distributed systems (Lamport and Lynch), networking, and security may interpret the work as a proposal for a convergence-oriented network where trust, identity, routing, and validation are co-designed rather than layered post hoc.

Readers seeking formal proofs, exhaustive physical modeling, or production-ready specifications should refer to the appendices or view this work as a conceptual and architectural foundation upon which such details may be developed.

Appendix **A** – Mathematical Formalism of Asynchronous Triangulation

- ❖ A.1 Motivation for a space-time-environment triangulation model
 - ❖ A.2 Formal definition of the triple integral formulation
 - ❖ A.3 Relation to Delaunay triangulation and dynamic mesh systems
 - ❖ A.4 Measurement constraints, bounds, and observability
 - ❖ A.5 Limitations and assumptions
-

Appendix **B** – Physical Considerations and Signal Interaction Models

- ❖ B.1 Photon-based signaling assumptions
 - ❖ B.2 Environmental interaction variables
 - ❖ B.3 Constraints on capture, disturbance, and detectability
 - ❖ B.4 Discussion of antiproton usage as a conceptual placeholder
 - ❖ B.5 Security implications and non-goals
-

Appendix C – Materials, Thermal Constraints, and Bandwidth Limits

- ❖ C.1 Silicon-based limitations
 - ❖ C.2 Graphene-based encapsulation and modulation
 - ❖ C.3 Thermal diffusion, electromigration, and longevity
 - ❖ C.4 Conservative performance bounds
 - ❖ C.5 References to prior experimental work
-

Appendix D – Identity Wrapping, CID Mapping, and On-Chain Anchoring

- ❖ D.1 Deterministic identifiers as network primitives
 - ❖ D.2 CID wrapping independent of ownership
 - ❖ D.3 Compatibility with public blockchains
 - ❖ D.4 Zero-knowledge alignment and privacy considerations
 - ❖ D.5 Threat model and adversarial behavior
-

Appendix E – Agent Models and Routing Heuristics

- ❖ E.1 Agent-on-Chip versus traditional SoC routing
- ❖ E.2 Cartographer agent path selection
- ❖ E.3 Validator agent integrity checks
- ❖ E.4 Latency, congestion, and feedback loops
- ❖ E.5 Failure modes

Introduction

1.1 The Problem Space: Secure Communication as a System Constraint

The global communication infrastructure relies on cryptographic assumptions that are increasingly misaligned with the long-term trajectory of computation. The anticipated maturation of large-scale quantum computing threatens widely deployed public-key systems, but the deeper issue extends beyond cryptography itself. Security, identity, routing, memory, and participation have historically been treated as separable layers, retrofitted onto networks that were not designed to withstand adversarial or post-quantum conditions.

This work adopts the position that secure communication in the quantum era is not a protocol-level problem, but a system-level constraint. Addressing it requires rethinking how devices enter a network, how trust is established, how keys are distributed, how data is routed, and how heterogeneous components coexist over time. Incremental cryptographic upgrades are insufficient if the surrounding architecture remains brittle, opaque, or centralized by construction.

1.2 The NEPO Project: Architectural Intent

The NEPO (Near Earth Proximity Orbit) Project proposes a modular, hybrid quantum-classical infrastructure designed to operate under long-term uncertainty in hardware capabilities, threat models, and scale. The intent is not to prescribe a single implementation, but to define an architectural framework in which security, scalability, and adaptability are treated as co-dependent primitives.

At its core, the NEPO architecture integrates quantum key distribution as an immutable security backbone while delegating computation, routing, and orchestration to classical systems augmented by specialized agents. Each component is designed to function independently yet remain composable, allowing the system to evolve without invalidating its foundational guarantees.

The project deliberately avoids assuming homogeneous devices, trusted ownership, or static network boundaries. Instead, it treats all participating elements—satellites, relays, consumer devices, and embedded systems—as accountable nodes within a unified but partitioned topology.

1.3 Scope, Non-Goals, and Design Philosophy

This document is intentionally architectural and declarative. It defines constraints, interfaces, and roles rather than complete mathematical proofs or physical implementations. Formal derivations, low-level signal mechanics, and hardware-specific optimizations are deferred to supplementary material or future work.

The guiding design philosophy is convergence without homogenization: heterogeneous systems are allowed to coexist, interoperate, and evolve while remaining verifiable and bounded. Modularity is treated not as an optimization, but as a requirement for longevity. The goal is to establish a framework that remains meaningful as quantum technologies mature, rather than one tightly coupled to current experimental limits.

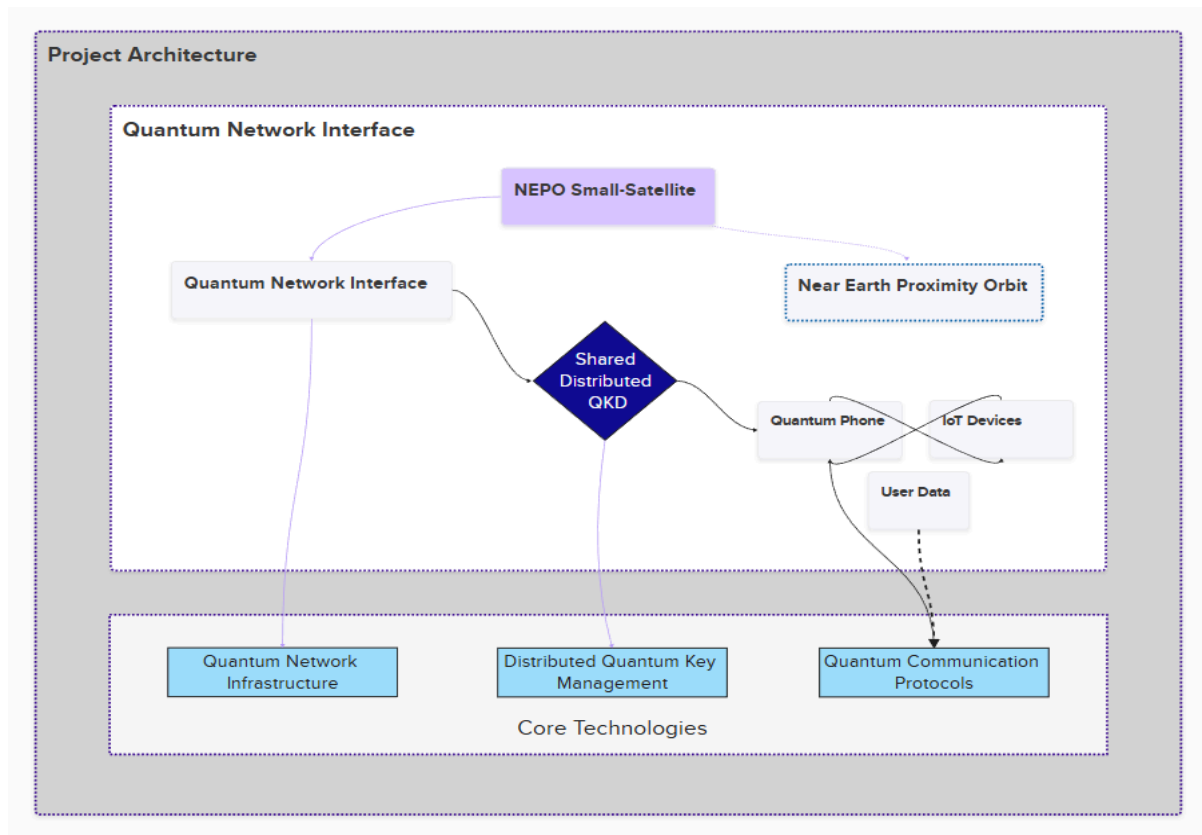
2. System Architecture and Network Model

2.1 A Two-Plane Network Model

The NEPO architecture is structured as a two-plane system:

1. **Infrastructure Plane** - Comprising space-based assets (near-Earth satellites), high-altitude or terrestrial relays, and backbone infrastructure responsible for long-range transmission, synchronization, and initial quantum key distribution.
2. **Participation Plane** - Comprising end-consumer devices, IoT and IIoT systems, and edge nodes that act simultaneously as endpoints and potential relays. Devices in this plane are not passive consumers but accountable participants in routing, validation, or resource sharing.

This separation allows the infrastructure plane to remain relatively stable while the participation plane scales organically. No single device is assumed to be trusted by default; accountability is enforced through deterministic identification and partitioned access.



2.2 Shared Distributed Quantum Key as a Structural Primitive

The Shared Distributed Quantum Key (QDK) functions as the cryptographic backbone of the NEPO system. Rather than serving as an isolated key exchange protocol, QDK is treated as a structural primitive embedded within the architecture.

Two operational modes are supported:

- *Singular Transmission Mode* - Point-to-point quantum key distribution for isolated, high-assurance communication paths.
- *Shared Partition Mode* - A distributed key framework enabling secure, multi-directional communication across a bounded set of participants.

By anchoring trust in physical properties governed by quantum mechanics, the system establishes a root of security that is independent of computational assumptions. Classical cryptographic mechanisms operate above this layer, inheriting its guarantees without replacing it.

2.3 Consumer Devices as First-Class Nodes

The NEPO framework introduces the concept of a "quantum phone" not as a specific product, but as a reference model for consumer-end participation. Such devices integrate reserved partitions for quantum key handling, classical computation, and agent-assisted routing within a layered hardware architecture.

This design ensures that end devices:

- Participate in secure key usage without managing quantum complexity directly
- Can act as transient relays or validators when permitted
- Remain accountable regardless of ownership or provenance

The distinction between infrastructure and user devices is therefore functional rather than hierarchical.

3. Conceptual Foundations and System Primitives

3.1 Convergence Through Asynchronous Topology

The NEPO system models the network as a dynamic, multi-layer mesh in which nodes are distributed across space, time, and operational context. Rather than assuming synchronous behavior or static topology, the architecture accommodates asynchronous triangulation among heterogeneous nodes.

This approach allows routing, validation, and key usage to adapt to latency, congestion, and environmental variation without requiring global synchronization. Mathematical formulations describing these interactions are used to define bounds and invariants, not to fully solve the system dynamics.

3.2 Deterministic Identity Wrapping

All devices and assets entering the network are assigned deterministic identifiers that function independently of ownership, trust, or prior registration. Identity wrapping is treated as a prerequisite for participation, not an optional feature.

This mechanism enables:

- Tagging and accountability of unknown or adversarial devices
- Decoupling identity from private data
- Compatibility with public or permissionless verification systems

The specific encoding formats and on-chain anchoring mechanisms are implementation-dependent and intentionally abstracted at this level.

3.3 Pixelized Validation Bits (PixBits) as a Validation and Control Surface

The Pixelized Validation Bits (pixBits) scheme functions as a shared validation metric rather than an encryption mechanism. It provides a visual and algorithmic representation of network state, routing directionality, and partition usage.

By mapping abstract system properties to immutable, non-interchangeable values, pixelized validation bits (pixBits) supports real-time verification, conflict prevention, and convergence monitoring across distributed components. Its role is to expose system invariants in a form accessible to both machines and operators.

3.4 Agents, Memory, and Routing Constraints

Specialized agents operate atop the physical hardware to manage routing, validation, mediation, and optimization. These agents interact with partitioned memory structures that distinguish between reserved, shared, and extensible regions.

Routing decisions are made locally and dynamically, informed by congestion, latency, and historical performance. This agent-assisted model reduces reliance on centralized coordination while preserving global coherence through shared constraints.

Sections beyond this point elaborate on hardware architecture, agent roles, cryptographic layering, and illustrative case studies. Detailed mathematical derivations, physical signal models, and encoding specifications are provided separately as supplementary material.

4. Hardware Architecture and Specialized Agents

4.1 Layered Hardware Architecture and Modularity

The NEPO hardware model is based on a layered, partition-aware microprocessor architecture designed to support heterogeneous workloads and long-term evolution. Rather than assuming a monolithic System-on-Chip (SoC), the architecture is decomposed into modular processing layers, each with explicit spatial, logical, and functional boundaries.

These layers may be implemented as stacked dies, adjacent boards, or distributed processing units, but are unified through deterministic interfaces and shared constraints. This design enables coexistence between classical computation, quantum key handling, routing logic, and agent-based orchestration without cross-contamination of trust domains.

The objective is not to maximize raw throughput in isolation, but to minimize systemic friction: latency, contention, thermal saturation, and architectural coupling.

4.2 Agent-on-Chip (AoC) as a Control Plane

Above the physical data plane operates an Agent-on-Chip (AoC) model. Agents are lightweight, specialized control entities embedded close to the hardware, operating within reserved partitions. They do not replace traditional computation; rather, they supervise routing, validation, scheduling, and optimization decisions in real time.

Unlike static routing tables or centralized controllers, AoC agents make decisions locally, based on instantaneous conditions such as congestion, signal quality, memory pressure, and timing constraints. This enables nanosecond-scale adaptation without external coordination.

4.3 Specialized Agent Roles

The architecture defines a bounded set of agent roles, each with a narrow and auditable scope:

- *Cartographer Agent*: Selects optimal routing paths based on latency, load, and environmental constraints.
- *Validator Agent*: Performs just-in-time integrity checks and consistency validation.
- *Mediator Agent*: Acts as an independent arbiter for transactions or state transitions requiring neutrality.
- *Security Operations Agent (SOC)*: Monitors anomalous behavior and enforces partition boundaries.
- *Processing and Encoding Agents*: Handle automation and format translation without accessing privileged state.

These agents form a cooperative but non-hierarchical mesh. No single agent has global authority; coherence emerges from shared constraints rather than centralized control.

5. Cryptographic and Security Model

5.1 Quantum Key Distribution as a Root of Trust

Quantum Key Distribution (Brassard and Bennett) (QKD) provides the foundational trust anchor of the NEPO system. By exploiting physical properties such as the no-cloning theorem (QueRa), QKD establishes shared secrets whose integrity is independent of computational hardness assumptions.

In NEPO, QKD is treated as a persistent structural layer rather than a transient protocol. Keys are generated, distributed, and consumed within bounded partitions, ensuring that compromise of higher layers cannot retroactively invalidate the root of trust.

5.2 Layered Classical Cryptography

Above the quantum layer, classical cryptographic mechanisms provide confidentiality, authentication, and integrity for data in transit and at rest. These mechanisms are modular and replaceable, allowing migration as algorithms age or are deprecated.

This layered approach explicitly avoids single points of cryptographic failure. Classical algorithms benefit from quantum-secured key material without being conflated with the quantum process itself.

5.3 Fault Tolerance and Integrity

Security in NEPO is reinforced through isolation, redundancy, and continuous validation. Each layer operates under a zero-trust assumption relative to adjacent layers. Faults—whether adversarial or environmental—are contained locally and surfaced through agent-mediated feedback loops.

This model aligns conceptually with principles from quantum error correction (Google): errors are detected and mitigated through structure and redundancy rather than assumption of perfect channels.

6. Illustrative Distribution Model

6.1 Conceptual Data Flow

To illustrate the architecture, the document introduces an abstract distribution model in which data propagates through layered partitions and directional paths. Visual and symbolic representations—such as directional markers and partition identifiers—serve as tools for reasoning about routing, access control, and convergence.

These representations are not implementation mandates but cognitive aids, enabling operators and systems to reason about state, flow, and constraints without exposing sensitive internals.

6.2 Feedback and Optimization Loop

The distribution model assumes a continuous feedback loop: system state is measured, deviations are identified, and corrective actions are applied through agent-mediated decisions. Residuals between expected and observed behavior drive reallocation of resources, rerouting, or throttling.

This control-loop perspective unifies routing, validation, and optimization under a single conceptual framework, independent of specific mathematical formulation.

7. Conclusion and Outlook

The NEPO Project proposes a foundational architecture for secure communication in the quantum era by treating security, identity, routing, and hardware constraints as inseparable design elements. Its strength lies not in any single component, but in the disciplined separation of concerns combined with rigorous composability.

By anchoring trust in quantum physical principles, enforcing deterministic identity, and delegating control to localized agents, the architecture is designed to remain adaptable under technological uncertainty. Mathematical formalisms, physical implementations, and material optimizations are intentionally deferred, preserving flexibility without sacrificing rigor.

This document serves as a declaration of architectural intent. It establishes boundaries, primitives, and invariants upon which future work—experimental, mathematical, and industrial—may be built. The ultimate measure of success is not immediate deployment, but longevity: the ability of the system to remain meaningful as computation, materials, and adversarial models evolve.

Furthermore;

8. Architectural Modularity, Constraints, and Deferred Formalism

8.1 Modularity as an Evolutionary Contract

The NEPO framework is grounded in a modular architectural philosophy in which the core engine and environmental constraints remain stable while adjacent components evolve over time. Modularity is treated as a contractual interface rather than a fixed implementation: successive versions of the system (e.g., v2, v3) may introduce new components, workflows, or external dependencies without invalidating prior guarantees.

In this context, versioning represents a form of architectural contract. Each iteration preserves compatibility at defined boundaries while allowing internal variability across microprocessing units, workflow orchestration, validation mechanisms, or external identity maintainers. External systems—such as decentralized identifier networks anchored to public ledgers—may serve as facilitators of verification without becoming trust authorities.

The objective of this unification is not maximal integration, but controlled interoperability across objects, processes, cells, systems, and dimensions. Convergence is therefore defined not as collapse into uniformity, but as the ability of heterogeneous elements to coordinate meaningfully under shared constraints.

8.2 Coexistence of Heterogeneous Environments

All units, extension types, data environments, and partitions—ranging from space-based infrastructure to the global ecosystem of IoT and IIoT devices—are assumed to coexist within the same network fabric. Safety is achieved through isolation, deterministic identification, and bounded interaction rather than exclusion.

Clusters may operate independently while still benefiting from shared services, keys, or validation primitives. This approach enables local autonomy without sacrificing global coherence and is explicitly designed to resist adversarial behavior through agility, redundancy, and compartmentalization.

Certain operational domains—such as baseband frequency allocation, multi-frequency signaling, antenna design, relay extension, or nanophotonic capture-and-retransmit mechanisms—are acknowledged as critical but are intentionally represented as independent variables. Their quantification, optimization, and long-term maintenance are deferred to specialized studies to avoid conflating architectural intent with hardware-specific tuning.

8.3 Materials Considerations and Comparative Constraints

Material choice imposes hard constraints on performance, longevity, and scalability. Silicon-based components remain effective and widely deployed; however, comparative research on graphene-based encapsulation demonstrates potential advantages in thermal conductivity, carrier mobility, and lifespan.

Rather than proposing novel materials science, NEPO adopts conservative comparative bounds derived from existing literature. Graphene's substantially higher thermal conductivity relative to silicon, reduced susceptibility to electromigration, and demonstrated high-frequency modulation capabilities motivate its consideration for relay and modulation components in high-density or long-lived deployments.

These considerations are treated as constraint-driven justifications, not performance guarantees. Material selection remains context-dependent and subject to empirical validation.

8.4 Bounded Formulation and Constraint Modeling

Where mathematical expressions are introduced, they serve to define bounds, relationships, and invariants rather than closed-form solutions. Variables such as maximum power output, relay contribution, frequency gating, and material coefficients are used to reason about feasibility and trade-offs.

Environmental factors—density, spatial configuration, and directional telemetry—are incorporated qualitatively to emphasize that routing and transmission are three-dimensional, adaptive processes rather than static planar abstractions. Formal coefficients may be defined locally within subsystems without requiring global resolution.

Detailed derivations, optimization models, and numerical simulations are intentionally omitted from the main body. Their inclusion would obscure the architectural narrative and prematurely constrain future implementations.

8.5 Asynchronous Spacetime Triangulation (Conceptual Overview)

The notion of asynchronous spacetime triangulation underpins routing and validation within the NEPO framework. Conceptually, the network is modeled as a dynamic mesh in which nodes occupy positions across space, time, and environmental context.

Mathematical constructs—such as multi-dimensional integrals inspired by triangulation methods—are employed as abstractions to reason about state convergence, path validation, and signal integrity. These constructs define what must be measurable and bounded, not how measurements are physically realized.

This abstraction allows the system to accommodate mobility, latency variation, and environmental interference without enforcing global synchronization. Formal mathematical treatment is deferred to supplementary work.

8.6 Security, Disclosure, and Omitted Detail

Certain technical details are deliberately omitted from this document. This is not to obscure the architecture, but to avoid disclosing sensitive operational, security, or implementation-specific information that could introduce risk or misinterpretation.

Omissions may include proprietary algorithms, detailed hardware schematics, low-level security protocols, unpublished experimental data, or vulnerability assessments. The absence of such details should be interpreted as a boundary of disclosure, not as a lack of rigor.

The architecture is presented in a manner intended to be examinable and dissectible without being exploitable. Supplementary materials and future publications may elaborate on specific components under appropriate disclosure conditions.

8.7 References and Prior Work

This work builds upon established results in quantum computing, quantum key distribution, error correction, and materials science. Publicly available references are cited for contextual grounding, while this document focuses on architectural synthesis rather than reproduction of existing results.

The contribution of NEPO lies in the integration and reframing of these domains into a coherent system-level architecture designed for longevity under technological uncertainty.

Works Cited

Brassard, Gilles, and Charles Bennett. "Bennett-Brassard Protocol."

<https://pages.vassar.edu/>, 1984,

<https://pages.vassar.edu/magnes/files/2018/12/BB84.pdf>. Accessed 8 5 2025.

Google. "Quantum error correction below the surface code threshold." *Google*

Scholar, 2025, *<https://www.nature.com/articles/s41586-024-08449-y>*.

Accessed 15 12 2025.

Lamport, Leslie, and Nancy Lynch. "Chapter on Distributed Computing." *Leslie*

Lamport, 3 February 1989,

<https://lamport.azurewebsites.net/pubs/lamport-chapter.pdf>. Accessed 17 December 2025.

QueRa. "What is No-Cloning Theorem." *QuEra Computing*, 2023,

<https://www.quera.com/glossary/no-cloning-theorem>. Accessed 17 December 2025.